

ME 300 Homework 3: Assumptions, Properties, and Ideal Gases

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

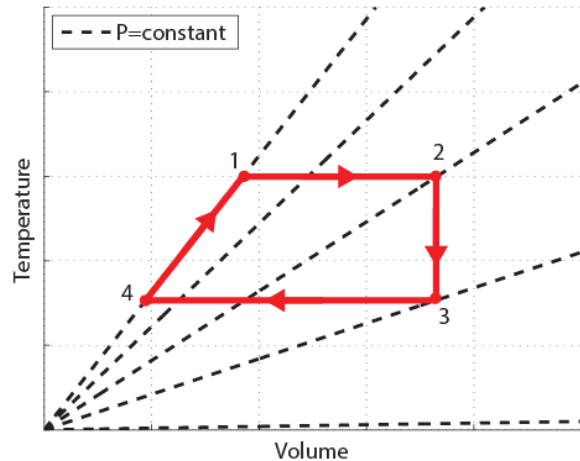
1. (20 points) A sealed cylindrical vessel has the inner dimensions of 58.2 inches tall and 9 inches in diameter. The vessel holds 25 lb of nitrogen (N_2) at 2900 psi.
 - Convert these specifications into SI units. What is the volume of the vessel in SI units?
 - How many moles of nitrogen are in the vessel?
 - What is the particular gas constant of nitrogen?
 - Assuming nitrogen is an ideal gas, what is the temperature of the nitrogen in the vessel?
 - Now I replace the nitrogen with the same number of moles of Argon at the same pressure. What is the temperature in the vessel now?
2. (10 points) A pressure transducer is hooked up to the vessel with N_2 (not N) in Problem #1, and the transducer reads gage pressure. Assume that the pressure from Problem #1 is the absolute pressure.
 - What is the gage pressure read by the transducer in SI units if the vessel is located in Washington DC (sea level)?
 - What happens to the gage pressure if I now take the vessel to Denver, CO?
3. (20 points) Below is a table of molecular weights of the most common substances we'll be using in this class. Use these molecular weights to answer the following questions.

Species	Molecular weight
Nitrogen (N_2)	28.0134 g/mol
Oxygen (O_2)	31.9988 g/mol
Carbon dioxide (CO_2)	44.01 g/mol
Water (H_2O)	18.01528 g/mol

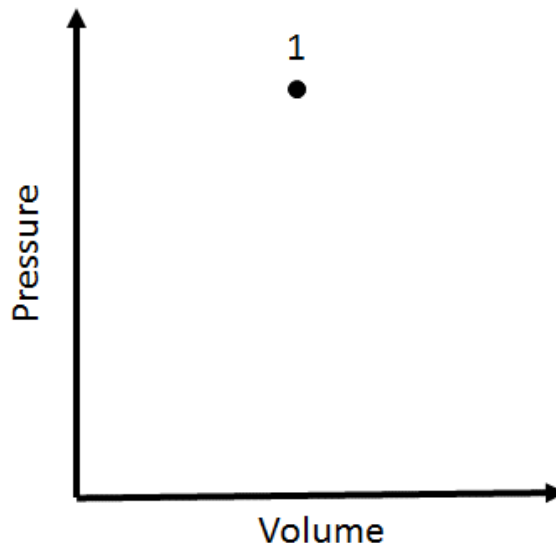
- a. What is the particular gas constant for O_2 ?
 - b. The mass-specific volume of a system of CO_2 is $10 \text{ m}^3/\text{kg}$ – what is its molar-specific volume?
 - c. Which substance has a particular gas constant of 188.92 J/kg-K ?
 - d. A gas cylinder has 20 kg of H_2O in it – how many moles of H_2O are in the cylinder?
 - e. The container has molar-specific volume of N_2 of $0.1 \text{ m}^3/\text{mol}$ – what is the density of this nitrogen?
4. (10 points) Explain the assumptions that must be met in order for a gas to be considered an ideal gas. What equation of state results from these assumptions?

5. (20 points) Consider the cycle shown in the T-v diagram, where isobars are shown in dotted lines, and answer the following questions assuming the working fluid is an ideal gas:
- a. Fill in the table below indicating whether each property increases, decreases, or is constant for each process:

Process	Temperature	Pressure	Volume
1-2			
2-3			
3-4			
4-1			



- b. Re-sketch this cycle on a P-v diagram, indicating the same state points and arrows for the cycle direction. I have provided the axes and State 1 for your reference.



6. (20 points) An ideal gas undergoes a four-step cycle with the following steps. Calculate the change in mass-specific internal energy and mass-specific enthalpy for each of the steps and then the total by summing up all the changes. Explain the answer you get for the total change in internal energy and enthalpy. Assume $c_v=700$ J/kg-K and $c_p=900$ J/kg-K.
- State 1 – State 2: Isothermal compression starting at $T_1=300$ K
 - State 2 – State 3: Isochoric heat addition to $T_3=800$ K
 - State 3 – State 4: Isothermal expansion
 - State 4 – State 1: Isochoric heat rejection